

Time-dependent density-functional theory for molecular photoionization with noniterative algorithm and multicenter *B*-spline basis set: CS₂ and C₆H₆ case studies

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In this work a new direct (noniterative) algorithm to solve the time-dependent density-functional theory equations for molecular photoionization has been proposed and implemented, using a multicentric basis set expansion of *B*-spline functions and complete exploiting of the molecular point-group symmetry. The method has been applied to study the photoionization dynamics of CS₂ and C₆H₆: the results confirmed the expectation of large screening effects in CS₂. For C₆H₆ the screening effects have been found to play a minor role than in CS₂, however, also in this case the quality of the final results is definitely improved. The method has proven suitable to study with confidence molecules of medium size, and there is still room for further improvement working on more elaborate treatment of the exchange-correlation functional. © 2005 American Institute of Physics. [DOI: 10.1063/1.1937367]

I. INTRODUCTION

In recent years density-functional theory (DFT) has found a very fertile field in quantum chemistry, where its applications have become practicable to large systems with affordable efforts. However, the validity of the conventional DFT, based on the Kohn–Sham scheme,¹ is limited to the electronic ground state (GS), hampering DFT applications to electron spectroscopy problems which involve electronic excited states.

On the other hand, the time-dependent density-functional theory (TDDFT) is a rigorous way to describe electronic excited states within the DFT framework.^{2,3} Pioneering TDDFT studies have been carried out about 25 years ago, both in the field of photoabsorption^{4–6} and dynamic polarizability,^{7,8} well before the formal assessment of the theory.⁹ More recently, the TDDFT implementations have been mainly directed to the fields of dynamic polarizability,^{10–12} hyperpolarizability,^{13–15} and discrete electronic excitations.^{12,16–21} Alternative implementations of the TDDFT method based on the resolution of the equations directly in the real space has also been proposed,^{22–26} and they have proven very powerful since with this approach molecules of medium size have been treated, for example, SiH₄ (Ref. 24) and cyclopropane.²⁶

Two different strategies are usually followed to solve the TDDFT equations employing finite basis set (the customary choice in quantum chemistry): with iterative (self-consistent) or direct (noniterative) algorithms. The former have been used to evaluate dynamic properties such as polarizabilities,¹⁰ the latter are employed to evaluate excitation energies and oscillator strengths.¹² Direct methods are

actually more general and can be used for both dynamical properties and excitations; a complete comparison between the two computational approaches is given in Ref. 21.

In this work we study the molecular photoionization at the TDDFT level, with a direct algorithm and a multicentric set of *B*-spline basis functions, with an explicit treatment of the continuum unbound wave function. The present scheme represents a twofold improvement with respect to our previous TDDFT molecular implementation:²⁷ first the basis set has been improved, from a crude one-center expansion (OCE) to a multicentric set, second the algorithm to solve the TDDFT equations, which was iterative in the original formulation of Ref. 27, is now direct. The benefits which arise from these choices are that the multicentric basis set allows to obtain convergent results with fewer basis functions than in the OCE approach, which is extremely slow to converge as the molecular size increases; moreover, the direct TDDFT algorithm avoids the difficulties of the self-consistent procedure, which sometimes has proven to be slow or even to fail to converge.

As test cases we have selected CS₂ and C₆H₆, because carbon disulphide contains two heavy sulphur nuclei which are expected to give rise to important response effects at TDDFT level while benzene is already a large molecule for such kind of calculations and represents a stringent test for numerical robustness, accuracy, and efficiency. Moreover CS₂ is a molecule whose photoionization has been studied both experimentally^{28–31} and theoretically,^{32,33} even recently due to its importance in molecular spectroscopy^{34–39} and astrophysics.^{40,41} C₆H₆ is an important system since it is the simplest aromatic compound, photoionization experiments on partial cross section of benzene are rather dated,⁴² however, an investigation on the photoelectron angular distribu-

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tion has recently appeared,⁴³ as well as accurate total photoabsorption⁴⁴ and photoionization⁴⁵ experiments and theoretical calculations.^{46–48}

II. THEORETICAL METHOD

Since TDDFT is an extension of the original Kohn–Sham (KS) formulation of the DFT theory and the output of the latter is the input of the former, it is convenient to shortly recall the basic KS theory. In the Kohn–Sham scheme¹ the following set of one-electron Schrödinger-like equations for n electrons are self-consistently solved:

$$H_{\text{KS}}\varphi_i = \varepsilon_i\varphi_i, \quad i = 1, \dots, n, \quad (1)$$

where

$$H_{\text{KS}} = -\frac{1}{2}\nabla^2 - \sum_N \frac{Z_N}{|\mathbf{r} - \mathbf{R}_N|} + \int \frac{n(\mathbf{r}')d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{XC}}[n] \quad (2)$$

is the usual KS Hamiltonian, n the electron density, Z_N the atomic numbers, $\mathbf{R}_N$ the nuclei coordinates, and V_{XC} the exchange-correlation potential.

In this work we employed the LB94 (Ref. 49) exchange-correlation potential, since its performances for both dynamic polarizability¹¹ and KS photoionization⁵⁰ have proven superior to local-density approximation (LDA) or generalized gradient approximation (GGA) conventional functionals, due to the correct asymptotic Coulomb behavior.

Once the self-consistent H_{KS} has been obtained, the continuum KS wave functions are extracted, that is the eigenfunctions of the H_{KS} with given positive eigenvalue equal to the photoelectron kinetic energy. The extraction of such unbound states can be recast according to Ref. 51 to the calculation of the eigenvectors with a minimum modulus eigenvalue of the energy-dependent matrix $\mathbf{A}^+\mathbf{A}$:

$$\mathbf{A}^+\mathbf{A}(E)c = ac, \quad \mathbf{A}(E) = \mathbf{H} - E\mathbf{S}, \quad (3)$$

where $\mathbf{H}$ and $\mathbf{S}$ are the Hamiltonian and overlap matrices, respectively, E the photoelectron kinetic energy, c are the eigenvectors, and a the minimum modulus eigenvalues. Such eigenvectors are efficiently obtained by block inverse iteration.⁵² In Eq. (3) a set of degenerate eigenvectors is extracted for each energy, the multiplicity of the set is equal to the number of the basis functions which obey to the continuum boundary conditions.

The TDDFT linear response of an electronic density to an external weak time-dependent electromagnetic field with frequency ω can be described, following Zangwill and Soven,⁴ by an effective self-consistent field (SCF) potential Φ^{SCF} , which consists of the external dipole potential Φ^{EXT} plus the Coulomb and exchange-correlation screening due to the redistribution of the electrons:

$$\Phi^{\text{SCF}}(\mathbf{r}, \omega) = \Phi^{\text{EXT}}(\mathbf{r}, \omega) + \int \frac{\delta n(\mathbf{r}', \omega)d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} + \left. \frac{\partial V_{\text{XC}}}{\partial n} \right|_{n(\mathbf{r})} \delta n(\mathbf{r}, \omega), \quad (4)$$

where in (4) $\delta n(\mathbf{r}, \omega)$ indicates the induced density and the further adiabatic local-density approximation,⁵³ (ALDA) of

the exchange-correlation kernel has been introduced. The induced density is obtained from the KS (noninteracting) dielectric susceptibility (χ) and the SCF potential:

$$\delta n(\mathbf{r}, \omega) = \int \chi(\mathbf{r}, \mathbf{r}', \omega) \Phi^{\text{SCF}}(\mathbf{r}', \omega) d\mathbf{r}'. \quad (5)$$

So the TDDFT formalism consists to solve the two coupled equations, (4) and (5), and different algorithms can be employed for this purpose. Iterative algorithms consist to start with the initial guess $\Phi^{\text{SCF}} = \Phi^{\text{EXT}}$ and then alternatively solve (5) and (4) until convergence is met for Φ^{SCF} . However, it is worth noting that the two equations are actually linear in Φ^{SCF} so the iterative procedure can be in principle avoided and a direct (noniterative) algorithm formulated. In fact, straightforward substitution of (5) in (4) gives the equation

$$\Phi^{\text{SCF}}(\mathbf{r}, \omega) = \Phi^{\text{EXT}}(\mathbf{r}, \omega) + \int \int K(\mathbf{r}, \mathbf{r}') \chi(\mathbf{r}', \mathbf{r}'', \omega) \Phi^{\text{SCF}}(\mathbf{r}'', \omega) d\mathbf{r}' d\mathbf{r}'', \quad (6)$$

where in (6) we have indicated with K the Coulomb and the ALDA exchange-correlation kernel:

$$K(\mathbf{r}, \mathbf{r}') = \frac{1}{|\mathbf{r} - \mathbf{r}'|} + \left. \delta(\mathbf{r} - \mathbf{r}') \frac{\partial V_{\text{XC}}}{\partial n} \right|_{n(\mathbf{r})}. \quad (7)$$

If now the integral Eq. (6) is solved with respect to Φ^{SCF} , the TDDFT problem can be recast avoiding the SCF procedure. Once Φ^{SCF} is obtained, all the parameters which describe the photoionization process, namely, the cross section and the asymmetry parameters, are calculated via transition moments using Φ^{SCF} instead of dipole operator in the conventional expressions.^{4,5}

III. COMPUTATIONAL IMPLEMENTATION

In this section we will consider in detail the multicentric basis set and the resolution of the direct TDDFT equation. Further technical details regarding the numerical integration schemes will be considered in the Appendix.

A. The multicentric basis set

The basis set choice for continuum calculations is not a trivial issue, due to the special requirements of the photoelectron wave function. Various choices have appeared in the literature, for example, with Lobatto functions^{46,54} or Coulomb waves.⁵⁵ The present choice is based on a multicentric linear combination of atomic orbital (LCAO) basis set of B -spline functions. Such choice has proven very promising, since it has been already implemented for KS continuum calculations⁵⁶ and its capabilities have been explored with excellent results in terms of accuracy and numerical stability. Therefore we have decided to employ the same approach to implement the present TDDFT scheme.

The LCAO basis set consists in a large OCE located on a chosen origin O :

$$\tau_{nlh\lambda\mu}^O = \frac{1}{r_O} B_n(r_O) X_{lh\lambda\mu}^0(\vartheta_O, \varphi_O) \quad (8)$$

supplemented by functions of the same type, which instead are symmetrized combinations of functions centered on the off-center arbitrary positions j :

$$\tau_{nlh\lambda\mu}^j = \sum_{j \in E_i} \frac{1}{r_j} B_n(r_j) X_{lh\lambda\mu}^{ij}(\vartheta_j, \varphi_j), \quad (9)$$

where i is an index which runs over the nonequivalent nuclei, E_i is the set of the equivalent nuclei, j runs over the equivalent nuclei and gives the origin of the off-center spherical coordinates, r represents the radial coordinate, B_n is the n th B -spline monodimensional function⁵⁷ while $X_{lh\lambda\mu}^0$ and $X_{lh\lambda\mu}^{ij}$ are the central and off-center (respectively) symmetry-adapted real spherical harmonics.⁵⁸ The symmetry-adapted real spherical harmonics consist of a linear combination of the real spherical harmonics $Y_{l,m}^R(\vartheta, \varphi)$ with proper coefficients $b_{lmh\lambda\mu}^0$ and $b_{lmh\lambda\mu}^{ij}$:

$$X_{lh\lambda\mu}^0(\vartheta, \varphi) = \sum_m Y_{lm}^R(\vartheta, \varphi) b_{lmh\lambda\mu}^0, \quad (10)$$

$$X_{lh\lambda\mu}^{ij}(\vartheta, \varphi) = \sum_m Y_{lm}^R(\vartheta, \varphi) b_{lmh\lambda\mu}^{ij}. \quad (11)$$

The indices have the following meaning: l and m are the usual angular momentum quantum numbers, λ indicates the irreducible representation (IR), μ the subspecies in the case of degenerate IR, and h is used to identify different elements with the same $\{l, \lambda, \mu\}$ set. The present implementation of the symmetry is general, that is also all nonabelian point groups are tractable, at variance with our previous work.⁵⁶

The B splines are built over the radial interval $[0, R_{\text{MAX}}^O]$ for the set $\{\tau_{nlh\lambda\mu}^O\}$ and over the intervals $[0, R_{\text{MAX}}^i]$ for the sets $\{\tau_{nlh\lambda\mu}^i\}$. The B splines are completely defined by their order and by the grids of knots for all the radial intervals $[0, R_{\text{MAX}}^O]$ and $[0, R_{\text{MAX}}^i]$. The off-center B -spline basis set is truncated: the last three functions are not included in the set in order to have the continuity of the second derivative over the surface of the spheres with radius R_{MAX}^i . In the present implementations the spheres of radius R_{MAX}^i cannot intersect each other. With this basis set, the Hamiltonian matrix $\mathbf{H}$ is partitioned in submatrices, as follows:

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_{00} & \mathbf{H}_{01} & \mathbf{H}_{0i} & \mathbf{H}_{0p} \\ \mathbf{H}_{10} & \mathbf{H}_{11} & 0 & 0 \\ \mathbf{H}_{i0} & 0 & \mathbf{H}_{ii} & 0 \\ \mathbf{H}_{p0} & 0 & 0 & \mathbf{H}_{pp} \end{pmatrix} \quad (12)$$

a block $\mathbf{H}_{00}$ between elements of the set $\{\tau_{nlh\lambda\mu}^O\}$, other blocks $\mathbf{H}_{ii}$ between the elements $\{\tau_{nlh\lambda\mu}^i\}$ belonging to the same i th center, and finally $\mathbf{H}_{0i}$ blocks between the sets $\{\tau_{nlh\lambda\mu}^O\}$ and $\{\tau_{nlh\lambda\mu}^i\}$. The details of the construction of the elements of the matrix blocks with a numerical technique are described in Ref. 56. Once the Hamiltonian and the overlap matrices are built, bound states are extracted by standard generalized diagonalization, and continuum orbitals by inverse iteration procedure according to Eq. (3). Bound and continuum orbitals are two ingredients necessary to calculate

dipole integrals, the third ingredient is the dipole potential which is trivial for KS case (x , y , and z components) while for TDDFT it must be calculated solving Eq. (6) by means of its representation in terms of the same LCAO B -spline basis set (8) and (9).

B. The resolution of the direct TDDFT equation

If Eq. (6) is represented over the LCAO B -spline basis set, we can indicate with $\mathbf{K}$ and χ the matrix representation of the kernel and dielectric susceptibility, respectively, with Φ^{SCF} and Φ^{EXT} the vector of the coefficients of the unknown SCF and the known external dipole potential, respectively. Solving with respect to Φ^{SCF} the following algebraic linear system is obtained:

$$(\mathbf{K}\chi - 1)\Phi^{\text{SCF}} = -\Phi^{\text{EXT}}. \quad (13)$$

In summary, if a method is given to build the two matrices which appear in (13) over the basis set, the TDDFT problem is recast to a linear set of algebraic equations, and the SCF procedure is avoided.

It is convenient to start with the construction of the matrix χ , which can be defined from Eq. (5) taking the scalar products with respect to basis elements:

$$\chi_{ij} = \langle \mu_i | \chi(\bar{r}, \bar{r}', \omega) | \lambda_j \rangle = \langle \mu_i | \delta n_j(\bar{r}, \omega) \rangle \quad (14)$$

in Eq. (14) indices i and j are the symbolic notations for the full set $\{nlh\lambda\mu\}$ which appears in definitions (8) and (9) for both centered and off-center basis elements, and the δn_j object designates the density induced by a potential equal to the j th basis element. The generic induced density δn_j can be obtained from the perturbation of the KS orbitals $\varphi_{ij}^{(1,\pm),3}$:

$$\delta n_j(\bar{r}, \omega) = \sum_i n_i (\varphi_i^* \varphi_{ij}^{(1,\pm)} + \varphi_i \varphi_{ij}^{(1,\mp)*}), \quad (15)$$

where n_i are the occupation numbers and $\varphi_{ij}^{(1,\pm)}$ are extracted from the solution of the inhomogeneous equation:

$$[H_{\text{KS}} - \varepsilon_i \pm \omega] \varphi_{ij}^{(1,\pm)} = -\hat{P} \tau_j \varphi_i. \quad (16)$$

In (16) the generic basis element τ_j has been substituted to the Φ^{SCF} of the inhomogeneous equation,³ ω indicates the photon energy, and $\hat{P}$ is a projector which orthogonalizes with respect to the occupied orbitals. Equation (16) can be represented in the LCAO basis set, obtaining a system of linear algebraic equations, where the left-hand side matrix is built from the unperturbed KS Hamiltonian and the overlap matrix, while the right-hand side consists in integrals between two basis elements and an occupied molecular orbital properly orthogonalized. More precisely there are as many different right-hand sides as basis elements for the potential identified by the j index run. So Eq. (16) can be solved with standard Lower-Upper (LU) decomposition for each photon energy. An amount of the homogeneous solution of Eq. (3) must be added to the so obtained unbound solution, in order to impose outgoing boundary conditions to the perturbed orbitals.³ Therefore the only new objects to be calculated to solve Eq. (16) are the integrals between the two basis elements and an occupied molecular orbital, such integrals are calculated with the same numerical integration scheme em-

ployed in the construction of the KS Hamiltonian, described in detail in Ref. 56. Although the calculation of these integrals is time consuming, they are independent with respect to the photon energy, therefore they are calculated at the beginning of the calculation, are saved on disk, and are read at each photon energy value, and in the general computational economy their calculation is a minimal fraction of the total cpu time if the number of energy points is sufficiently high. This procedure avoids the numerical calculation of new integrals at each cycle necessary for the iterative procedure, which hampers the efficiency of iterative algorithms as in our previous OCE implementation,²⁷ this problem has been also found in the standard TDDFT quantum chemistry codes where it has been observed that the calculation of the SCF potential over the grid points represents the most consuming step in the calculation.⁵⁹ Once Eq. (16) has been solved, the coefficients of $\varphi_{ij}^{(1,\pm)}$ are obtained, and the last step needed to get the elements of the dielectric susceptibility matrix (14) is to perform a scalar product:

$$\chi_{ij} = \langle \mu_i | \delta n_j(\vec{r}, \omega) \rangle = \sum_k n_k (\langle \mu_i | \varphi_k^* \varphi_{kj}^{(1,\pm)} \rangle + \langle \mu_i | \varphi_k \varphi_{kj}^{(1,\mp)*} \rangle) \quad (17)$$

the scalar products in the last right-hand side of Eq. (17) are easily calculated performing a matrix multiplication between the matrix of the $\varphi_{ij}^{(1,\pm)}$ coefficients and the integrals between one orbital and two basis functions, exactly the same objects which appear in the right-hand side of Eq. (16) and this completes the construction of the dielectric susceptibility matrix. The next step is the construction of the kernel matrix $\mathbf{K}$ which, at variance of χ , is energy independent and therefore is computed at the beginning of the calculation and stored on disk for further usage. Such matrix operates over the density scalar products and gives the coefficients of the potential generated by the given density, as the sum of a Coulomb and an exchange-correlation part, $\Phi^{\text{SCF}} = \Phi^{\text{C}} + \Phi^{\text{XC}}$. The Coulomb coefficients Φ^{C} can be efficiently extracted solving the Poisson equation in the same basis set: $\nabla^2 \Phi^{\text{C}} = -4\pi \delta n_j$, so they are related to the density scalar products by matrix multiplication with the inverse of the Laplacian matrix:

$$\Phi^{\text{C}} = -4\pi (\nabla^2)^{-1} \delta n_j, \quad (18)$$

where in (18) $(\nabla^2)^{-1}$ is the inverse of the Laplacian matrix and δn_j the vectors whose elements are the scalar products (17). Also the exchange-correlation response Φ^{XC} can be represented in the given basis with coefficients Φ^{XC} and recast in a matrix-vector product, in fact, $\Phi^{\text{XC}} = \mathbf{Z} \delta n_j$, with $\mathbf{Z} = \mathbf{M}^{-1}$, and the elements of $\mathbf{M}$ are

$$M_{ij} = \langle \tau_i | 1/\partial n_{(0)}^{(0)} V_{\text{XC}} | \tau_j \rangle. \quad (19)$$

This step involves only the inversion of the $\mathbf{M}$ matrix, while in our previous scheme two products by the inverse $\mathbf{S}$ matrix were required.²⁷ Although the methods are equivalent only for complete basis set, we have chosen the more economic since numerical tests showed negligible differences.

Finally, the elements of the $\mathbf{K}$ matrix are therefore

$$K_{ij} = -4\pi (\nabla^2)_{ij}^{-1} + Z_{ij} \quad (20)$$

this step completes the construction of all the objects necessary appearing in Eq. (13), which is solved by standard LU decomposition technique employing the Lapack library. Once the solution vector Φ^{SCF} of the coefficients is obtained, the integrals between an initial occupied orbital, the SCF potential and the continuum orbital, necessary to calculate the cross section and the asymmetry parameters, are calculated from the integrals between the occupied orbitals and the two basis functions, performing two summations running over both continuum and SCF potential coefficients.

IV. COMPUTATIONAL DETAILS

For CS₂ the bond length of 2.93 a.u. and for C₆H₆ the C–C and C–H distances of 2.64 and 2.0485 a.u., respectively, are employed in the calculations, taken from the experimental equilibrium geometry.

The self-consistent electron density given by the ADF (Refs. 60 and 61) program and used in the successive LCAO *B*-spline steps is obtained with the LB94 (Ref. 49) exchange-correlation potential employing the ground-state electronic configuration and a STO DZP basis set.

The origin of the OCE is set on the center of mass of both the molecules. The radial *B*-splines basis set is of order 10, and is built on a linear grid of step size 0.2 a.u. for both molecules with $R_{\text{MAX}}^{\text{O}} = 20$ a.u., additional knots have been added near the off-center nuclei positions. The expansion on the off-center nuclei is divided with a radial linear grid with a step size of 0.2 a.u. up to values of $R_{\text{MAX}}^{\text{I}} = 1.4$ a.u. for S in CS₂ and with a step size of 0.2 a.u. up to values of $R_{\text{MAX}}^{\text{I}} = 1.2$ a.u. and $R_{\text{MAX}}^{\text{I}} = 0.6$ a.u. for C and H, respectively, in C₆H₆.

The highest angular momentum employed in the OCE is indicated as $L_{\text{MAX}}^{\text{O}}$, and in the calculations we have employed $L_{\text{MAX}}^{\text{O}} = 16$ for both CS₂ and C₆H₆. The highest angular momentum employed in the off-center spheres around the nuclei positions is $L_{\text{MAX}}^{\text{I}} = 2$ for S in CS₂, $L_{\text{MAX}}^{\text{I}} = 1$ for C, and $L_{\text{MAX}}^{\text{I}} = 0$ for H in C₆H₆.

In order to have a better comparison with the experiment and to smooth the sharp structures of Feshbach resonances, the cross-section profiles are convoluted with a Gaussian function with full width at half maximum (FWHM) of 1 eV, and the experimental ionization thresholds have been used instead of KS eigenvalues in Eq. (16).

Finally, in order to give a quantitative idea of the computational effort actually needed for such kind of calculations, we report the maximum dimension of the Hamiltonian matrices and the central processing unit (CPU) time needed to calculate the cross section at a given photon energy, for runs on a HP Itanium Server RX2600 professional workstation: for CS₂ dim=714 and CPU $\cong$ 20 s, while for C₆H₆ dim=3127 and CPU $\cong$ 1900 s, with effort scaling approximately with cubic power of dimension due to the linear algebraic system nature of the problem.

TABLE I. KS eigenvalues of the CS₂ orbitals calculated with the ADF and with the B-spline LCAO methods. Experimental ionization potentials are also reported. All values in eV.

Orbital	ADF	B-spline LCAO			Expt.
		$L_{\text{MAX}}^o = 10$	$L_{\text{MAX}}^o = 16$	$L_{\text{MAX}}^o = 20$	
$1\sigma_g$	-2444.135	-2444.189	-2444.131	-2444.130	
$2\sigma_g$	-292.796	-292.934	-292.934	-292.933	293.1
$3\sigma_g$	-224.486	-224.535	-224.493	-224.493	234.2
$4\sigma_g$	-170.712	-170.940	-170.891	-170.892	170.03
$5\sigma_g$	-25.649	-25.687	-25.690	-25.690	25.2
$6\sigma_g$	-17.237	-17.259	-17.261	-17.262	16.19
$1\sigma_u$	-2444.136	-2444.189	-2444.131	-2444.130	
$2\sigma_u$	-224.487	-224.524	-224.493	-224.493	234.2
$3\sigma_u$	-170.712	-170.921	-170.890	-170.892	170.03
$4\sigma_u$	-23.512	-23.559	-23.566	-23.567	20.17
$5\sigma_u$	-15.112	-15.185	-15.184	-15.184	14.47
$1\pi_g$	-170.573	-170.730	-170.750	-170.753	170.03
$2\pi_g$	-11.454	-11.460	-11.485	-11.485	10.06
$1\pi_u$	-170.573	-170.686	-170.749	-170.752	170.03
$2\pi_u$	-14.305	-14.458	-14.465	-14.465	12.83

V. RESULTS AND DISCUSSION

A. CS₂

In order to assess the basis set requirements for this molecule, we have performed a preliminary set of test calculations with different values of the maximum angular momentum of the expansion on the origin L_{MAX}^o , the KS eigenvalues have been reported in Table I for $L_{\text{MAX}}^o = 10, 16$, and 20 , together with the ADF values and the experimental ionization energies for comparison. It is found that convergence is quite fast, since differences between $L_{\text{MAX}}^o = 10$ and 16 are below 0.1 eV and between $L_{\text{MAX}}^o = 16$ and 20 are even around few meV. Therefore we have selected $L_{\text{MAX}}^o = 16$ in order to be sure to have convergent results. The same convergence test has been applied to both cross section and asymmetry parameter KS profiles (not reported), and we have found in practice a complete convergence for $L_{\text{MAX}}^o = 16$, which has been employed in all further calculations at both KS and TDDFT levels.

The cross section and the asymmetry parameter profiles relative to the four outermost valence orbitals of CS₂ are reported in Figs. 1–4, calculated at both TDDFT and KS levels together with the available experimental data. Starting with the highest occupied molecular orbital (HOMO) $2\pi_g$, the cross-section results at low photon energy (below 20 eV) are considered in the left upper panel of Fig. 1. The KS curve displays a smooth decreasing behavior, while TDDFT profile is much more intense than the former and carries several features associated with autoionization resonances, which are less and less intense as the energy increases. The experimental data²⁹ are in good agreement with the TDDFT profile for energy above 13 eV; near the threshold the accord is less quantitative, although the experimental point at 12 eV suggests the presence of a sharp structure probably not resolved by the limited experimental resolution, such structure is found to slightly lower energy in the TDDFT profile. It is worth noting that the autoionization resonances cannot be

described at all at the simpler KS level, therefore the TDDFT level of theory has the double advantage to give more quantitative results and to describe new phenomenology. In the right upper panel the high-energy region up to 120 eV is considered, where a Cooper minimum is found in both KS and TDDFT curves. In this case the TDDFT curve is in excellent agreement with the experimental data,^{29,31} reproducing with high accuracy the rapid decay before the Cooper minimum, around which the experimental data are rather

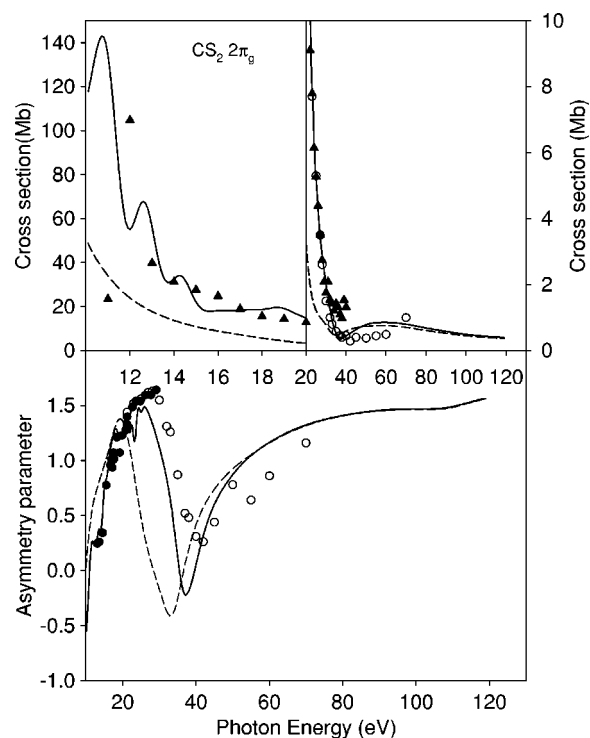


FIG. 1. The cross section (upper panels) and the asymmetry parameter (lower panel) of the $2\pi_g$ orbital of CS₂. Solid line: TDDFT results, dashed line: KS results, ●: experiment Ref. 30, ○: experiment Ref. 31, and ▲: experiment Ref. 29.

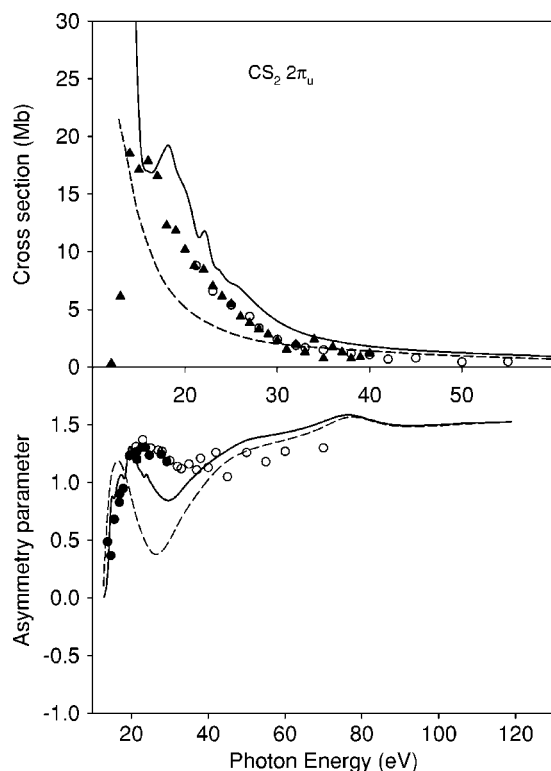


FIG. 2. The cross section (upper panel) and the asymmetry parameter (lower panel) of the $2\pi_u$ orbital of CS_2 . Solid line: TDDFT results, dashed line: KS results, $\bullet$: experiment Ref. 30, $\circ$: experiment Ref. 31, and $\blacktriangle$: experiment Ref. 29.

scattered. In the lower panel of Fig. 1 the asymmetry parameter is considered: the calculated curves display the common waving feature typical of atomic $3p$ orbitals, and also in this case the TDDFT results are in much better agreement with respect to the experiment than the KS ones. A global shift to higher energy is observed passing from the KS to the TDDFT level of theory, and even the rather complex threshold behavior is nicely reproduced by the TDDFT calculation.

The next $2\pi_u$ orbital is considered in Fig. 2: the TDDFT cross-section profiles is in better agreement with respect to the experimental data than the KS curve, although the calculated intensity is slightly overestimated. The asymmetry parameter is also rather sensitive to the method: at least for energy below 40 eV the KS and the TDDFT curves show different behavior in the description of the observed minimum. Such minimum is exaggerated in the KS profile while the shallow shape given at the TDDFT level is in much better agreement with respect to the experiment. In any case the threshold behavior is nicely reproduced.

In the upper panel of Fig. 3 the cross-section profiles of the $5\sigma_u$ orbital are reported: the TDDFT results are in excellent agreement with the experimental data, and in this case it is very noticeable the improvement with respect to the KS formalism, which gives a profile which decays too fast. The asymmetry parameter (lower panel) displays a rather complicated shape: the TDDFT curve carries a first structure with two very narrow maxima, followed by a second broader structure which ends with a smooth tail. The comparison with the experiment is also in this case very nice: the agreement is in practice quantitative with the exception of the

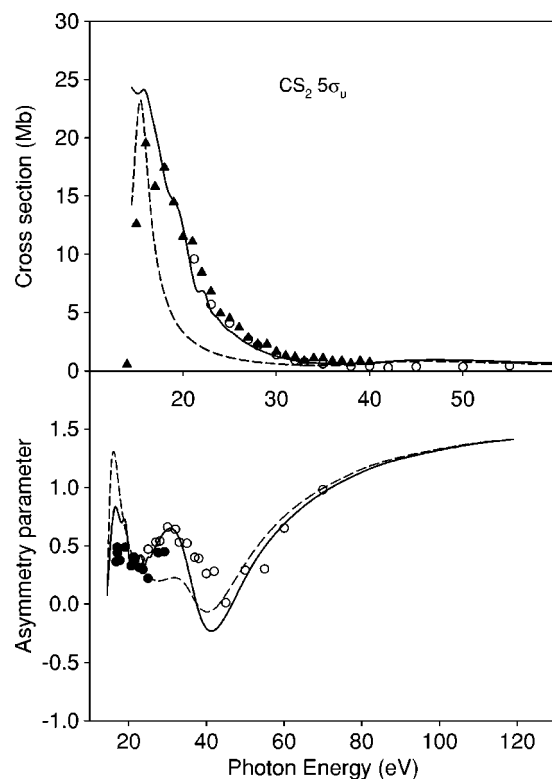


FIG. 3. The cross section (upper panel) and the asymmetry parameter (lower panel) of the $5\sigma_u$ orbital of CS_2 . Solid line: TDDFT results, dashed line: KS results, $\bullet$: experiment Ref. 30, $\circ$: experiment Ref. 31, and $\blacktriangle$: experiment Ref. 29.

narrow threshold feature and with minimal deterioration elsewhere. It is quite interesting to observe that in this specific case the KS results fail to describe the second broad structure around 30 eV, which appears just sketched.

Finally, the results concerning the $6\sigma_g$ orbital are considered in Fig. 4. Also in this case the difference between the KS and the TDDFT profiles is rather important at low photon energy, although above 35 eV the curves become almost superimposed. In this case the comparison with the experiment is difficult, since the two sets of experimental data do not match each other and therefore the situation is uncertain. Moreover it should be noticed that from the experimental photoelectron spectrum³¹ it has been found that the main line is followed by a rather strong satellite structure, which may withdraw intensity from the main line; this effect is not included in the TDDFT formalism and is probably the reason of the deterioration of the quality of the TDDFT results in the inner valence region, where satellite typically are rather intense. In the asymmetry parameter the differences between the KS and the TDDFT results are not dramatic, and despite the agreement with the experimental data is not quantitative, the global shape of the measured curve is properly reproduced by the theory.

In summary we can say that for CS_2 the TDDFT results compare very well with respect to the experiment, for both the cross-section and the asymmetry parameter profiles of the four outer valence ionizations. Moreover the differences with respect to the simpler KS method are quite noticeable, since the KS method gives always too low intensity in the cross

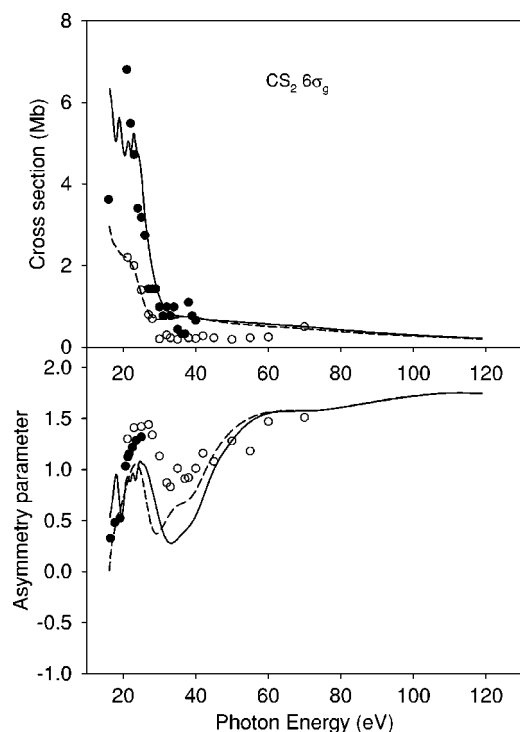


FIG. 4. The cross section (upper panel) and the asymmetry parameter (lower panel) of the $6\sigma_g$ orbital of CS_2 . Solid line: TDDFT results, dashed line: KS results, $\bullet$: experiment Ref. 30, $\circ$: experiment Ref. 31, and $\blacktriangle$: experiment Ref. 29.

section and shifts the profiles towards the threshold. Such observed differences between the methods suggests that the screening effects, that is the response of the electron density to the external time-dependent electromagnetic field, is rather pronounced in CS_2 . This finding is not surprising due to the presence of heavy S atoms, in fact, electrons of second row elements are more mobile than those of first row ones. Higher electron mobility makes heavy elements more polarizable, increasing the importance of the screening effects.

B. C_6H_6

The same preliminary test calculations already described for CS_2 have been performed for C_6H_6 as well, the KS eigenvalues have been reported in Table II for $L_{\text{MAX}}^0=10$, 16, and 20, together with the ADF values and the experimental ionization energies for comparison. Also in this case convergence is nicely met already for $L_{\text{MAX}}^0=16$, and further analysis of both cross-section and asymmetry parameter KS profiles (not reported) indicates complete convergence for $L_{\text{MAX}}^0=16$, which has been employed in all further calculations at both KS and TDDFT levels.

In Figs. 5 and 6 the cross-section and the asymmetry parameter profiles of C_6H_6 valence orbitals are reported, calculated at both KS and TDDFT levels together with the available experimental data.

Starting with the HOMO $1e_{1g}$ in Fig. 5, the TDDFT profile shows important difference with respect to the KS results, in particular, the very intense threshold intensity and a strong feature with a maximum at 14.1 eV. These new features must be ascribed to autoionization Feshbach reso-

TABLE II. KS eigenvalues of the C_6H_6 orbitals calculated with the ADF and with the B -spline LCAO methods. Experimental ionization potentials are also reported. All values in eV.

Orbital	ADF	B -spline LCAO			Expt.
		$L_{\text{MAX}}^0=10$	$L_{\text{MAX}}^0=16$	$L_{\text{MAX}}^0=20$	
$1a_{1g}$	-290.530	-290.568	-290.536	-290.536	290.38
$2a_{1g}$	-25.862	-25.835	-25.858	-25.862	25.9
$3a_{1g}$	-17.849	-17.673	-17.889	-17.925	16.9
$1e_{1g}$	-11.588	-11.696	-11.782	-11.784	9.2
$1e_{2g}$	-290.525	-290.557	-290.524	-290.524	290.38
$2e_{2g}$	-19.588	-19.494	-19.590	-19.605	19.2
$3e_{2g}$	-13.092	-12.931	-13.141	-13.170	11.5
$1a_{2u}$	-14.185	-14.327	-14.442	-14.445	12.3
$1b_{1u}$	-290.523	-290.548	-290.520	-290.520	290.38
$2b_{1u}$	-16.108	-15.653	-16.007	-16.059	15.5
$1b_{2u}$	-15.564	-15.659	-15.819	-15.822	14.8
$1e_{1u}$	-290.530	-290.560	-290.532	-290.532	290.38
$2e_{1u}$	-23.103	-22.997	-23.074	-23.086	22.5
$3e_{1u}$	-15.128	-14.788	-15.149	-15.198	13.9

nances which are missing in the KS profile since it cannot be described at that level of theory. The agreement with the experiment is not quantitative, since this new feature is actually present in the experimental profile but falls at higher energy, namely, around 15.5–18.0 eV with a discrepancy of few eVs. By the way it is quite encouraging that the experimental feature, at least qualitatively, can be reproduced by theory and clearly ascribed to autoionization resonances. Of course it would be interesting to identify the possible causes of the quantitative deteriorations about the energy position of the observed features. We reasonably attribute this difficulty to deficiencies of the LB94 exchange-correlation potential employed in the resolution of the KS Eq. (1). This hypothesis is suggested by a systematic error found when moderate large size molecules have been considered already at the KS level, for example, $\text{Cr}(\text{C}_6\text{H}_6)_2$ (Ref. 62) or C_{60} (Ref. 63), where KS eigenvalues have been found too bound with respect to ionization potentials and profiles shifted too much towards the threshold, if compared with respect to molecules of smaller size.^{64,65} This finding suggests that there is a need to test the performances of more elaborate exchange-correlation functional than LB94 in the TDDFT calculations on medium-sized systems to identify an optimal choice. The asymmetry parameter profile of the $1e_{1g}$ orbital is in good agreement with the experimental data, although in this case the differences between KS and TDDFT methods are rather small. The next $3e_{2g}$ orbital is also considered in Fig. 5, in this case the TDDFT curve matches better the experiment than the KS one, since it reduces the too high KS intensity at threshold, although the maximum is shifted to lower energy, as observed and discussed for the $1e_{1g}$ orbital. The $3e_{2g}$ TDDFT asymmetry parameter reproduces correctly the shape of the measured data, with the exception of the region around 18–23 eV where too low values are calculated.

In the two central panels of Fig. 5 the results relative to the $1a_{2u}$ orbitals are considered, with evidence of large deviations with respect to the experiment for both cross-section

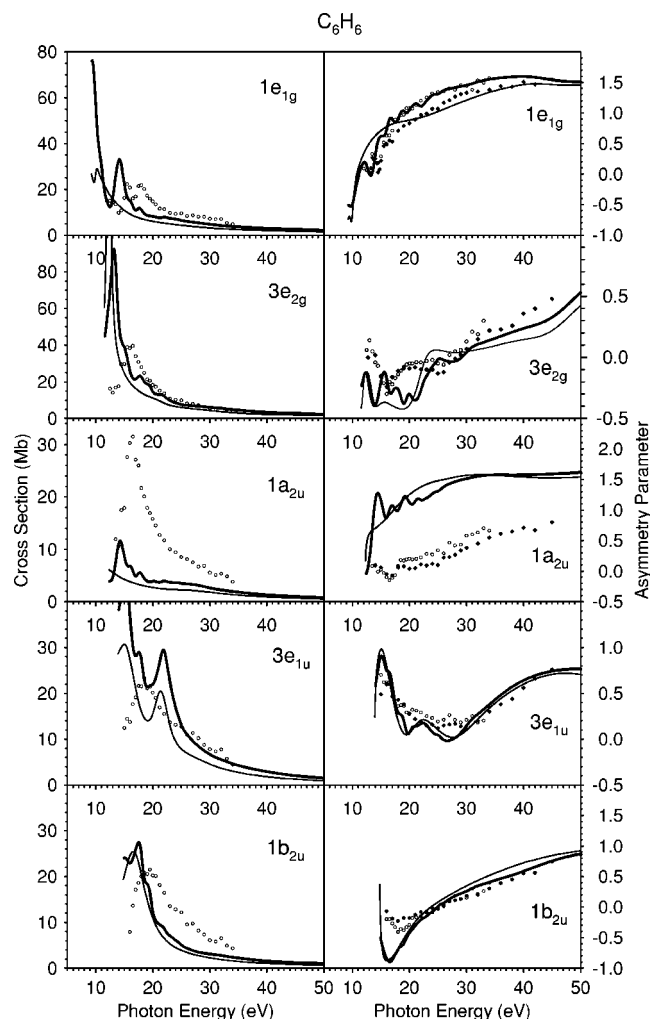


FIG. 5. The cross section and the asymmetry parameter profiles of the $1e_{1g}$, $3e_{2g}$, $1a_{2u}$, $3e_{1u}$, and $1b_{2u}$ orbitals of C_6H_6 . Thick line: TDDFT results, thin line: KS results. Experimental data: $\circ$ Ref. 42, $\blacklozenge$ Ref. 43.

and asymmetry parameter profiles. Since there are no specific arguments to expect a complete deterioration of the performance of the method for specific orbitals we are led to ascribe the disagreement between the theory and the experiment to the latter. In fact, it is well apparent from the photoelectron spectrum reported in Ref. 42 that $1a_{2u}$ and $3e_{2g}$ overlap considerably, therefore it might be expected that errors may be likely introduced during the band deconvolution.

The next $3e_{1u}$ orbital (Fig. 5) shows moderate response effects in the cross section, the TDDFT giving higher intensity than the KS, while negligible effects are found in the asymmetry parameter, which follows very closely the experimental data.

For the orbitals $1b_{2u}$ (lower panel of Fig. 5) and $2b_{1u}$ (upper panel of Fig. 6), in general the differences between the TDDFT and the KS are rather modest, suggesting a minor role played by response effects. Interestingly, while for $1b_{2u}$ the calculated cross section is too low with respect to the experiment, for $2b_{1u}$ the opposite behavior is found. The asymmetry parameter profiles of both orbitals are in fairly nice agreement with the measured data.

In Fig. 6 the results relative to $3a_{1g}$ and $2e_{2g}$ orbitals are also reported: for $3a_{1g}$ cross section the agreement with the

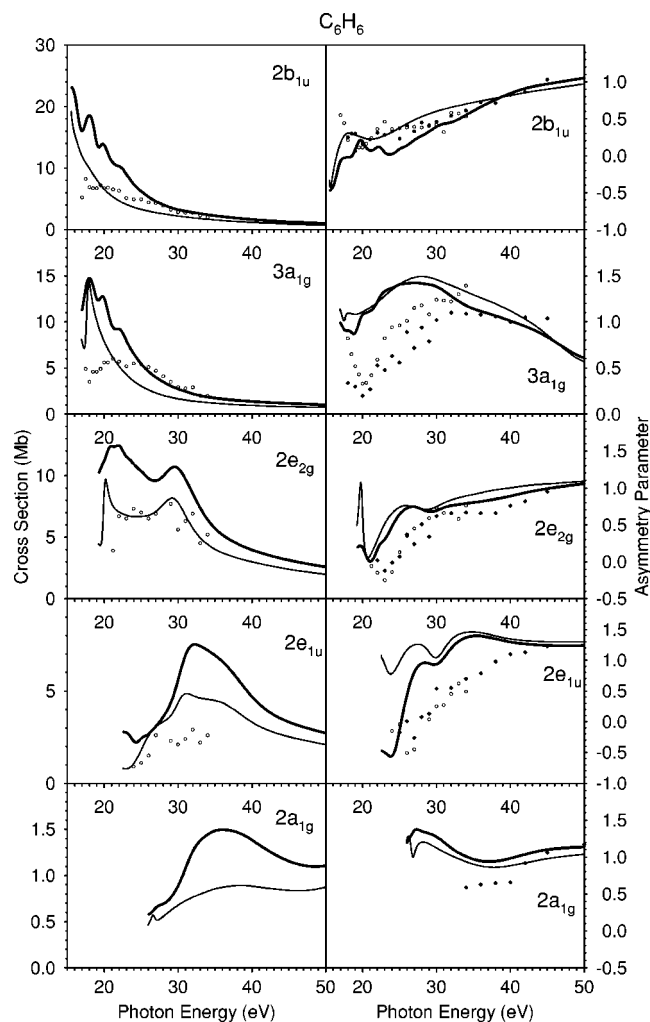


FIG. 6. The cross section and the asymmetry parameter profiles of the $2b_{1u}$, $3a_{1g}$, $2e_{2g}$, $2e_{1u}$, and $2a_{1g}$ orbitals of C_6H_6 . Thick line: TDDFT results, thin line: KS results. Experimental data: $\circ$ Ref. 42, $\blacklozenge$ Ref. 43.

experiment is not good since too high intensity is calculated near the threshold by both methods. The $3a_{1g}$ asymmetry parameter is instead in better agreement with the measured data, in fact, despite the low-energy region, the decreasing behavior at high energy is properly reproduced by the method, as it will be considered in more detail further when the asymmetry parameters will be analyzed up to 120 eV. The next $2e_{2g}$ orbital instead shows better agreement between theory and experiment for the asymmetry parameter, although for the cross section the TDDFT formalism seems to worsen with respect to the KS one.

In the lower panels of Fig. 6 the $2e_{1u}$ and $2a_{1g}$ orbitals are considered, in this case the experimental data are rather sparse and, more important, their inner valence nature suggests that a deterioration of the results may be expected due to the presence of satellite structures which subtract intensity to the cross section as a consequence of shake-up process.

In Fig. 7 we have reconsidered the asymmetry parameter profiles of all the orbitals, but now up to 120 eV in the photon energy scale, in order to compare the results with a recent measurement.⁴³ As a general remark, the KS and TDDFT profiles show only tiny differences at high energy, while near the threshold the two curves tend to differentiate each other.

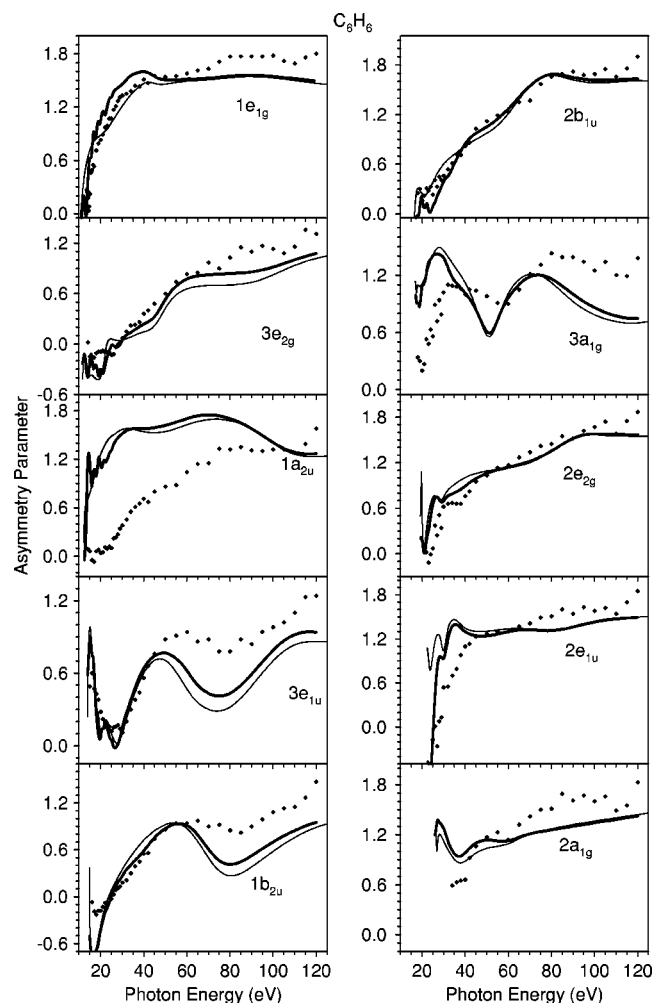


FIG. 7. The asymmetry parameter profiles relative to all valence orbitals of C_6H_6 up to 120 eV in the photon energy scale. Thick line: TDDFT results, thin line: KS results. Experimental data: $\blacklozenge$ Ref. 43.

For all the orbitals considered the agreement with respect to the experiment is excellent, when small deviations are observed like, for example, in $3e_{1u}$, $1b_{2u}$, and $3a_{1g}$ the global curve shape is always well reproduced. The only exception, with very large deviations with respect to the experiment, is the $1a_{2u}$ orbital, and as already discussed this may be possibly ascribed to the deconvolution errors of experimental data.

Finally, in Fig. 8 the total cross section of C_6H_6 calculated at both KS and TDDFT levels is reported, together with very recent photoabsorption⁴⁴ and photoionization⁴⁵ experimental data. The TDDFT curve displays a rather large complex feature in the range of 12–26 eV, rich of structures which originate from the opening of the different ionization edges as well as from the structures present in the individual photoionization profiles. The TDDFT curve is qualitatively similar to the KS one, but is in much better agreement with the experimental data since it is generally more intense especially for energies above 20 eV. A noticeable feature is the suppression of the huge resonance obtained at about 13 eV in the KS results, redistributing this intensity at higher energy. It is quite interesting to analyze in more detail the tails of the theoretical and experimental data at high-energy val-

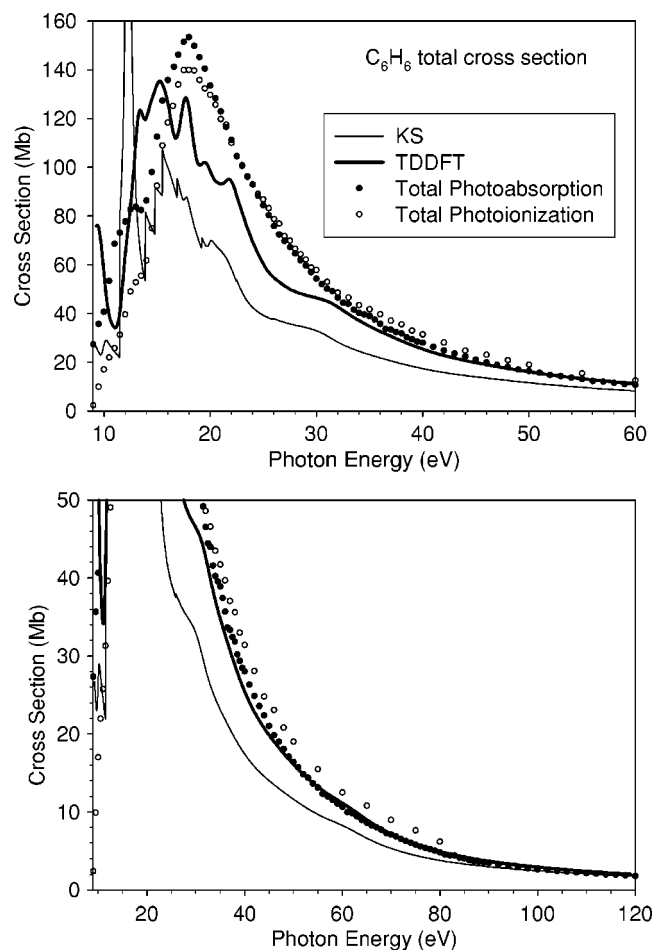


FIG. 8. The total cross section of C_6H_6 . Thick line: TDDFT results, thin line: KS results. Experimental data: $\bullet$ Ref. 44, $\circ$ Ref. 45.

ues, namely, up to 120 eV, in the lower panel of Fig. 8. The TDDFT curve gives an excellent reproduction of the decaying behavior of the experimental data, recovering the already small deviation found in the KS curve; so in general we may say that the TDDFT performs quite well in the high-energy region, while for lower energies the agreement is less quantitative but still much better than that obtained at the KS level.

We may trace the following conclusions about the TDDFT results of C_6H_6 . In general the TDDFT method performs better than the KS one in terms of the agreement with the experimental data, in some instances it introduces new features which are absent at the KS level (autoionization resonance in HOMO ionization). However, the response effects are not dramatic, at least if compared with the results of CS_2 . Moreover the agreement between theory and experiment is found to be better for the asymmetry parameter than for the cross section, and this may be attributed to the quality of the measured data, which are recent for the former⁴³ and rather dated for the latter.⁴² There is one case where the disagreement is quite large, namely, the $1a_{2u}$ orbital, and this has been attributed to errors introduced during the deconvolution procedure of the experimental data. The TDDFT method with multicentric B -spline basis with present direct algorithm implementation has therefore proven very efficient

and numerically stable to treat systems of medium size such as benzene with relatively modest computational efforts.

VI. CONCLUSIONS

In this work a new direct algorithm to solve the TDDFT equations for molecular photoionization has been proposed and implemented, using a multicentric basis set expansion of *B*-spline functions and complete exploitation of the molecular point-group symmetry. Such algorithm is direct in the sense that it avoids the self-consistent procedure usually employed in the TDDFT codes, and this makes it more efficient because there is no need to calculate numerical integrals in each step of the SCF procedure and is numerically more robust because iterative algorithms may suffer convergence problems. The method has been applied then to study the photoionization dynamics of CS₂ and C₆H₆; carbon disulphide has been chosen because of the presence of two heavy sulphur atoms which are expected to increase the importance of the screening effects included in the TDDFT formalism, while benzene has been considered because it still remains a rather large system for continuum calculations and therefore it represents a stringent case to test the performance of the method in terms of efficiency and stability. The results confirmed the expectation of large screening effects in CS₂, in fact, the TDDFT results so obtained are in much better agreement with the experiment than the KS ones. For C₆H₆ the screening effects have been found to play a minor role than in CS₂, however, also in this case the quality of the final results is definitely improved. The deviations with respect to the experiment, although strongly reduced, are still present and are attributed to possible deficiencies of the LB94 exchange-correlation potential. Therefore the method is suitable to study with confidence molecules of medium size, and there is still room for further improvement working on more elaborate treatment of the exchange-correlation functional.

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APPENDIX: DETAILS OF THE NUMERICAL INTEGRATION SCHEMES

In the description of the method and its implementation it has been underlined that matrix elements have been built employing the numerical method already developed in our previous KS LCAO *B*-spline method of Ref. 56. However, the TDDFT scheme requires also special integrals between an occupied orbital and two basis elements, whose construction deserves special care in order to gain high accuracy. The molecular orbital is calculated from its basis set expansion, so the integrand function is the product of three functions which may be centered all on the origin, all on an off-centered nucleus or some can lie on the origin and some on one off-center nucleus. No more than one off-center nucleus

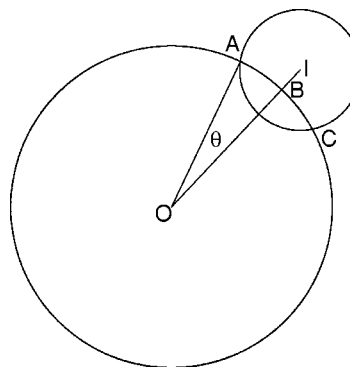


FIG. 9. The geometrical scheme employed for the numerical integration between two basis functions on the origin and the off-center contribution of the molecular orbital, see text for details (Appendix).

may be involved in the integration because in the present implementation atomic spheres are not allowed to intersect each other. Different schemes must be employed according to the location of the centers of the functions involved.

- Three functions all on the origin: this case is trivial, since the angular part becomes analytical and numerical integration over the radial coordinate is easily and efficiently performed by standard Gauss–Legendre method.
- When at least one function lies on one of the off-center nuclei: a local off-center spherical reference system is introduced with polar axis through origin, and integration is performed numerically along radial and polar coordinates and analytically along the azimuthal coordinate, as described in detail in Ref. 56, but with the exception considered in the next point:
- It has been noted that when basis functions lie on the origin and the integrals with the off-center part of the molecular orbital are needed, a more accurate scheme must be employed. We have found that a very accurate and efficient scheme consists to develop the off-center part of the molecular orbital in a radial expansion centered on the origin for each l, m angular momentum contribution, and then to integrate along the radial coordinate like in the first case (three functions all on the origin). The numerical effort consists to calculate the integrals between the off-center molecular orbital contribution and the real spherical harmonic centered on the origin for each radial knot. The geometry of such scheme is sketched in Fig. 9, where the circle centered on the origin *O* corresponds to a specific value of the radial coordinate, and the circle centered on *I* corresponds to a generic off-center sphere. It is sufficient to integrate over the spherical sector defined by the rotation of the arc *ABC* around the *OI* axes, employing a bidimensional numerical Gauss–Legendre quadrature along the polar θ angle *BA* and along the 2π complete azimuthal angle around the *OI* axes.

In order to exploit complete point group symmetry, previous integrals are calculated only for nonequivalent atoms, because the contribution of all the other equivalent atoms can be easily obtained simply by operating a rotation of the non-

equivalent integrals and, in the case of nonabelian groups, a linear combination of the degenerate subspecies. Integral rotation is a simple and economic procedure since the angular basis functions employed are real spherical harmonics and rotation matrix are analytically calculated from Euler angles.⁶⁶

- ¹W. Kohn and L. J. Sham, Phys. Rev. **140**, A1133 (1965).
- ²M. E. Casida, in *Recent Developments and Applications of Modern Density Functional Theory*, Theoretical and Computational Chemistry Vol. 4 edited by J. M. Seminario, (Elsevier Science, Amsterdam, 1996), p. 391.
- ³G. D. Mahan and K. R. Subbaswamy, *Local Density Theory of Polarizability* (Plenum, New York, 1990).
- ⁴A. Zangwill and P. Soven, Phys. Rev. A **21**, 1561 (1980).
- ⁵Z. H. Levine and P. Soven, Phys. Rev. A **29**, 625 (1984).
- ⁶Z. H. Levine, Phys. Rev. A **36**, 1181 (1987).
- ⁷M. J. Stott and E. Zaremba, Phys. Rev. A **21**, 12 (1980).
- ⁸G. D. Mahan, Phys. Rev. A **22**, 1780 (1980).
- ⁹E. Runge and E. K. U. Gross, Phys. Rev. Lett. **52**, 997 (1984).
- ¹⁰S. J. A. van Gisbergen, J. G. Snijders, and E. J. Baerends, J. Chem. Phys. **103**, 9347 (1995).
- ¹¹S. J. A. van Gisbergen, V. P. Osinga, O. V. Gritsenko, R. van Leeuwen, J. G. Snijders, and E. J. Baerends, J. Chem. Phys. **105**, 3142 (1996).
- ¹²C. Jamorski, M. E. Casida, and D. R. Salahub, J. Chem. Phys. **104**, 5134 (1996).
- ¹³S. J. A. van Gisbergen, J. G. Snijders, and E. J. Baerends, J. Chem. Phys. **109**, 10644 (1998).
- ¹⁴S. J. A. van Gisbergen, J. G. Snijders, and E. J. Baerends, Phys. Rev. Lett. **78**, 3097 (1997).
- ¹⁵S. J. A. van Gisbergen, J. G. Snijders, and E. J. Baerends, J. Chem. Phys. **109**, 10657 (1998).
- ¹⁶R. Bauernschmitt and R. Ahlrichs, Chem. Phys. Lett. **256**, 454 (1996).
- ¹⁷R. Bauernschmitt, M. Häser, O. Treutler, and R. Ahlrichs, Chem. Phys. Lett. **264**, 573 (1997).
- ¹⁸K. B. Wiberg, R. E. Stratmann, and M. J. Frisch, Chem. Phys. Lett. **297**, 60 (1998).
- ¹⁹R. E. Stratmann, G. E. Scuseria, and M. J. Frisch, J. Chem. Phys. **109**, 8218 (1998).
- ²⁰S. J. A. van Gisbergen, F. Kootstra, P. R. T. Schipper, O. V. Gritsenko, J. G. Snijders, and E. J. Baerends, Phys. Rev. A **57**, 2556 (1998).
- ²¹A. Görling, H. H. Heinze, S. Ph. Ruzankin, M. Staufer, and N. Rösch, J. Chem. Phys. **110**, 2785 (1999).
- ²²X.-M. Tong and S.-I. Chu, Phys. Rev. A **55**, 3406 (1997).
- ²³K. Yabana and G. F. Bertsch, Phys. Rev. A **60**, 3809 (1999).
- ²⁴T. Nakatsukasa and K. Yabana, J. Chem. Phys. **114**, 2550 (2001).
- ²⁵I. Vasiliev, S. Ögüt, and J. R. Chelikowsky, Phys. Rev. B **65**, 115416 (2002).
- ²⁶T. Nakatsukasa and K. Yabana, Chem. Phys. Lett. **374**, 613 (2003).
- ²⁷M. Stener and P. Decleva, J. Chem. Phys. **112**, 10871 (2000).
- ²⁸G. R. Cook and M. Ogawa, J. Chem. Phys. **51**, 2419 (1969).
- ²⁹F. Carnovale, M. G. White, and C. E. Brion, J. Electron Spectrosc. Relat. Phenom. **24**, 63 (1981).
- ³⁰T. A. Carlson, M. O. Krause, F. A. Grimm, J. D. Allen, Jr., D. Mehaffy, P. R. Keller, and J. W. Taylor, J. Chem. Phys. **75**, 3288 (1981).
- ³¹T. A. Carlson, M. O. Krause, and F. A. Grimm, J. Chem. Phys. **77**, 1701 (1982).
- ³²R. E. Stratmann and R. R. Lucchese, J. Chem. Phys. **97**, 6384 (1992).
- ³³R. E. Stratmann and R. R. Lucchese, J. Chem. Phys. **101**, 9548 (1994).
- ³⁴C. Y. R. Wu and D. L. Judge, J. Chem. Phys. **78**, 2180 (1983).
- ³⁵S. Kakar, H.-C. Choi, and E. D. Poliakov, J. Chem. Phys. **97**, 4690 (1992).
- ³⁶P. Baltzer, B. Wannberg, M. Lundqvist *et al.*, Chem. Phys. **202**, 185 (1996).
- ³⁷D. Aitchison and J. H. D. Eland, Chem. Phys. **263**, 449 (2001).
- ³⁸T. Masouka, A. Okaji, and A. Kobayashi, Int. J. Mass. Spectrom. **226**, 317 (2003).
- ³⁹G. J. Rathbone, E. D. Poliakov, J. D. Bozec, and R. R. Lucchese, J. Chem. Phys. **122**, 064308 (2005).
- ⁴⁰F. Z. Chen and C. Y. R. Wu, Geophys. Res. Lett. **22**, 2131 (1995).
- ⁴¹R. V. Yelle and M. A. McGrath, Icarus **119**, 90 (1996).
- ⁴²T. A. Carlson, P. Gerard, M. Krause, F. A. Grimm, and B. P. Pullen, J. Chem. Phys. **86**, 6918 (1987).
- ⁴³P. Baltzer, L. Karlsson, B. Wannberg, G. Öhrwall, D. M. P. Holland, M. A. MacDonald, M. A. Hayes, and W. von Niessen, Chem. Phys. **224**, 95 (1997).
- ⁴⁴R. Feng, G. Cooper, and C. E. Brion, J. Electron Spectrosc. Relat. Phenom. **123**, 199 (2002).
- ⁴⁵R. Feng, G. Cooper, and C. E. Brion, J. Electron Spectrosc. Relat. Phenom. **123**, 211 (2002).
- ⁴⁶I. Wilhelmy, L. Ackermann, A. Görling, and N. Rösch, J. Chem. Phys. **100**, 2808 (1994).
- ⁴⁷M. Venuti, M. Stener, and P. Decleva, Chem. Phys. **234**, 95 (1998).
- ⁴⁸M. Stener, S. Furlan, and P. Decleva, Phys. Chem. Chem. Phys. **3**, 19 (2001).
- ⁴⁹R. Van Leeuwen and E. J. Baerends, Phys. Rev. A **49**, 2421 (1994).
- ⁵⁰M. Stener, S. Furlan, and P. Decleva, J. Phys. B **33**, 1081 (2000).
- ⁵¹C. Froese Fischer and M. Idrees, Comput. Phys. **3**, 53 (1989).
- ⁵²M. Brosolo and P. Decleva, Chem. Phys. **159**, 185 (1992).
- ⁵³E. Gross and W. Kohn, Adv. Quantum Chem. **21**, 255 (1990).
- ⁵⁴N. Rösch and I. Wilhelmy, Chem. Phys. Lett. **189**, 499 (1992).
- ⁵⁵V. P. Osinga, B. C. Oostenbrink, E. van Lenthe, C. F. Guerra, and E. J. Baerends, Chem. Phys. **284**, 565 (2002).
- ⁵⁶D. Toffoli, M. Stener, G. Fronzoni, and P. Decleva, Chem. Phys. **276**, 25 (2002).
- ⁵⁷C. de Boor, *A Practical Guide to Splines* (Springer, New York, 1978).
- ⁵⁸P. G. Burke, N. Chandra, and F. A. Gianturco, J. Phys. B **5**, 2212 (1972).
- ⁵⁹S. J. A. van Gisbergen, J. G. Snijders, and E. J. Baerends, Comput. Phys. Commun. **118**, 119 (1999).
- ⁶⁰E. J. Baerends, D. E. Ellis, and P. Ros, Chem. Phys. **2**, 41 (1973).
- ⁶¹C. Fonseca Guerra, J. G. Snijders, G. te Velde, and E. J. Baerends, Theor. Chem. Acc. **99**, 391 (1998).
- ⁶²M. Stener, G. Fronzoni, S. Furlan, and P. Decleva, J. Chem. Phys. **114**, 306 (2001).
- ⁶³P. Colavita, G. De Alti, G. Fronzoni, M. Stener, and P. Decleva, Phys. Chem. Chem. Phys. **3**, 4481 (2001).
- ⁶⁴M. Stener, G. Fronzoni, D. Toffoli, and P. Decleva, Chem. Phys. **282**, 337 (2002).
- ⁶⁵G. Fronzoni, M. Stener, and P. Decleva, J. Chem. Phys. **118**, 10051 (2003).
- ⁶⁶A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, Princeton, New Jersey, 1974).